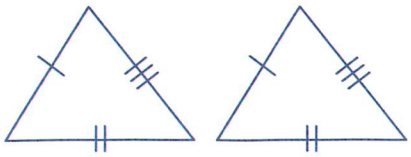
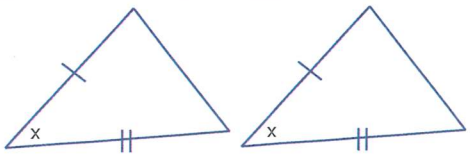
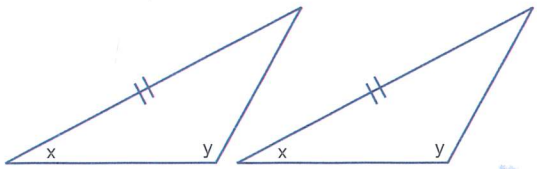
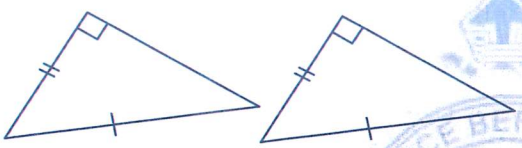
Grade 10 Congruency Questions

SUMMARY OF THE 4 CASES FOR CONGRUENCE OF TRIANGLES

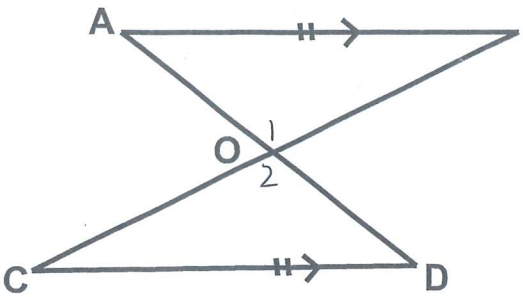
1. 	SSS = SSS
2. 	SAS = SAS (included angle)
3. 	AAS = AA corresponding S
4. 	RHS = RHS

Note: Triangles are ***not necessarily*** congruent if:

1. a a a = a a a
2. s s a = s s a — For congruence the angle MUST BE the INCLUDED angle or a RIGHT ANGLE
3. a a s = a a s — For congruence the sides MUST CORRESPOND, i.e. they must be opposite EQUAL ANGLES

QUESTION 1

Prove that $AO = OD$ and $BO = OC$.



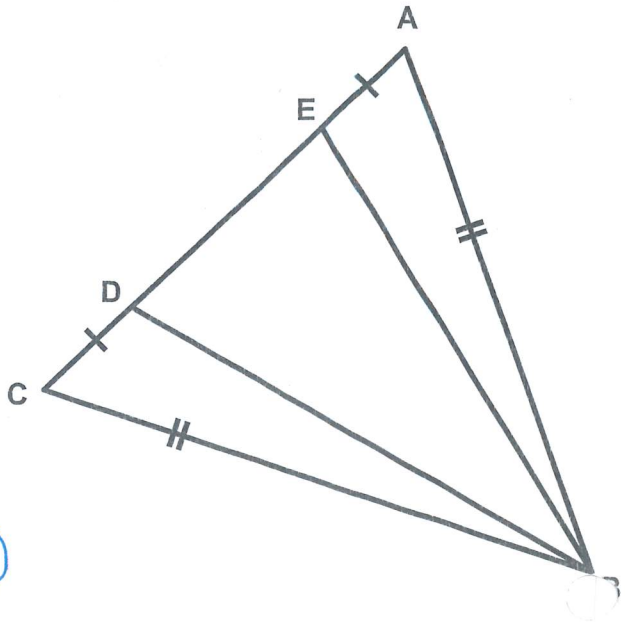
In $\triangle AOB$ & $\triangle DOC$

1. $AB = CD$ (Given)
2. $\hat{A} = \hat{D}$ (alt $\angle$s : AB // CD)
3. $\hat{O}_1 = \hat{O}_2$ (Vert opp $\angle$s)

$\therefore \triangle AOB \equiv \triangle DOC$ (SAA)

$\therefore$ Deduce: $AO = OD$, $BO = OC$ } sides of $\equiv \triangle$s

QUESTION 2



a) What type of triangle is ΔABC?

Isosceles because $AB = BC$

b) Prove that $\Delta ABE \equiv \Delta CBD$.

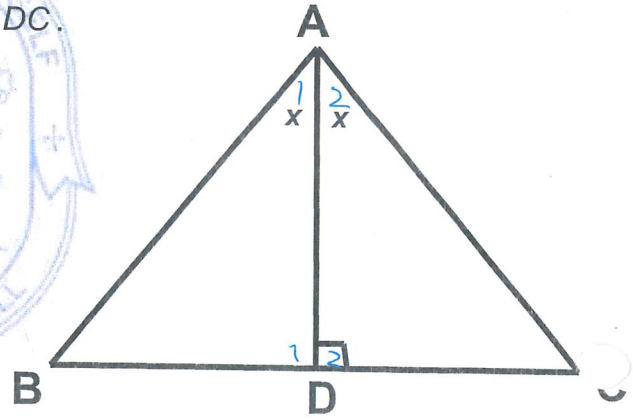
In Δ ABE & Δ CBD

1. $\hat{A} = \hat{C}$ (isos Δ; $AB = BC$)
2. $AE = DC$ (given)
3. $AB = BC$ (given)

$\Delta ABE \equiv \Delta CBD$ (SAS)

QUESTION 3

Prove that $AB = AC$ and $\hat{B} = \hat{C}$ and $BD = DC$.



In Δ ABD & Δ ACD

1. $\hat{A}_1 = \hat{A}_2 = x$ (given)
2. AD is common
3. $\hat{D}_1 = \hat{D}_2 = 90°$ (str line)

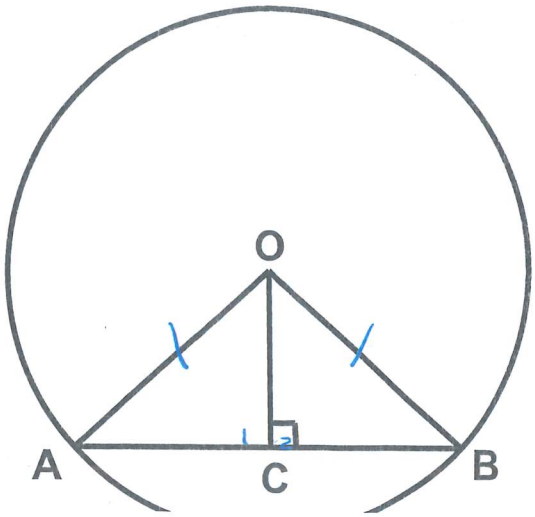
$\therefore \Delta ABD \equiv \Delta ACD$ (SAA)

$AB = AC$ $\quad \hat{B} = \hat{C}$ } deduced

$BD = DC$

QUESTION 4

O is the centre of the circle. Prove that $AC = CB$.



In Δ AOC & BOC

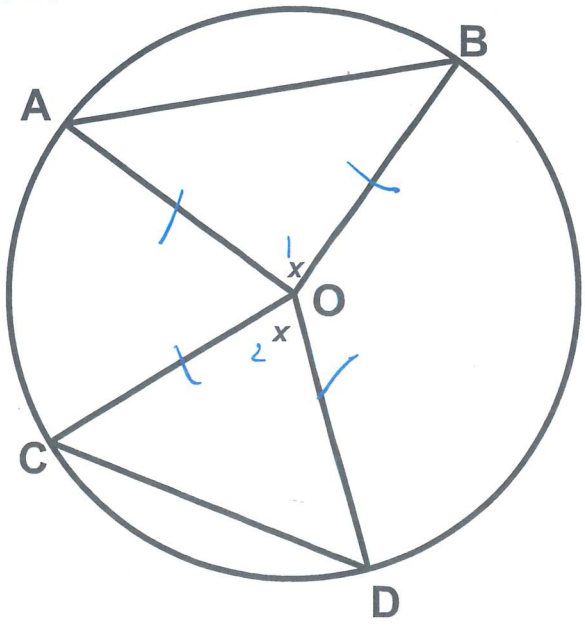
1. AO = BO (radii of circle)
2. OC is common
3. $\hat{C}_1 = \hat{C}_2 = 90°$ (str line)

$\Delta AOC \equiv \Delta BOC$ (RHS)

Deduce AC = CB (sides of $\equiv$ Δs)

QUESTION 5

O is the centre of the circle. Prove that $AB = CD$.



In $\triangle ABO$ & CDO

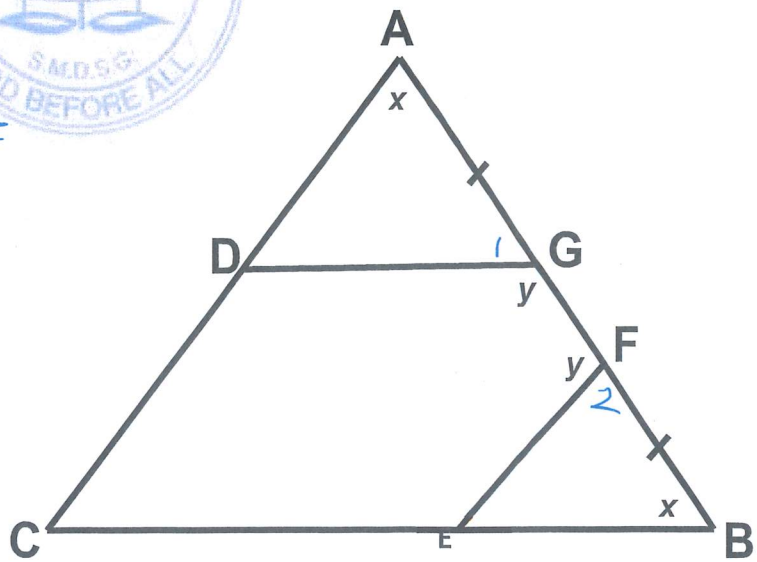
1. $AO = OC$ } radii
2. $BO = OD$ }
3. $\hat{O}_1 = \hat{O}_2 = x$ (given)

$\triangle ABO \equiv \triangle CDO$ (SAS)

$AB = CD$ (deduced)

QUESTION 6

Prove that $AD = BE$.



In $\triangle ADG$ & $\triangle BEF$

1. $\hat{A} = \hat{B} = x$ (given)
2. $AG = BF$ (given)
3. $\hat{G}_1 = 180° - y$ } adj $\angle$s on str lines
   $\hat{F}_2 = 180 - y$ }

$\therefore \hat{G}_1 = \hat{F}_2$

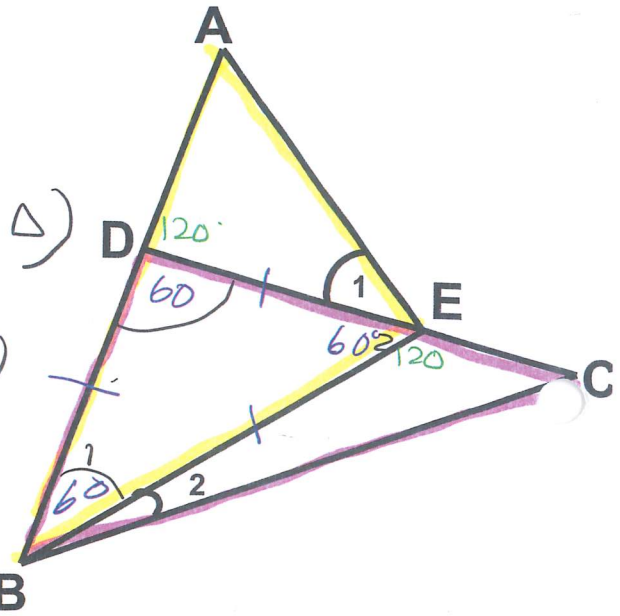
$\triangle ADG \equiv \triangle BEF$ (SAA)

$AD = BE$ (sides of $\equiv \triangle$s)

QUESTION 7

In the diagram ΔDBE is an equilateral triangle. $\hat{E}_1 = \hat{B}_2$

Prove that: a) $\Delta ABE \equiv \Delta CDB$.



In ΔABE & ΔCDB

1. $\hat{E}_1 = \hat{B}_2$ (given)

$\hat{B}_1 = \hat{E}_2 = 60°$ (∠s of equilateral Δ)

$\therefore \hat{E}_1 + \hat{E}_2 = \hat{B}_1 + \hat{B}_2$ ΔDBE

2. $\hat{A} = \hat{C}$ (remaining ∠s of Δs)

b) $AD = EC$

3. $BE = BD$ (equilateral Δ)

$\Delta ABE \equiv \Delta CDB$ (SAA)

b) RTP: $AD = EC$

In ΔADE & ΔCEB

1. $\hat{A} = \hat{C}$ proved
2. $DE = BE$ sides of equilateral Δ
3. $\hat{E}_1 = \hat{B}_2$ given

LOR

$AB = CD$ sides of $\equiv \Delta$s

$AD + BD = EC + DE$

But $BD = DE$ Equilateral Δ

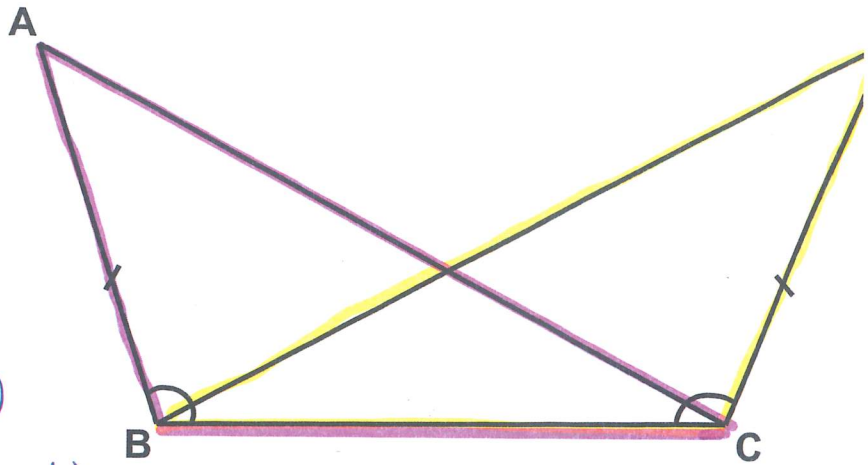
$\therefore AD = EC$

$\Delta ADE \equiv \Delta CEB$ (SAA)

$AD = EC$ (deduced)

QUESTION 8

If $AB = DC$ and $A\hat{B}C = D\hat{C}B$, prove that $AC = DB$.



In ΔABC & ΔDCB

1. $AB = DC$ given
2. BC is common
3. $A\hat{B}C = D\hat{C}B$ given

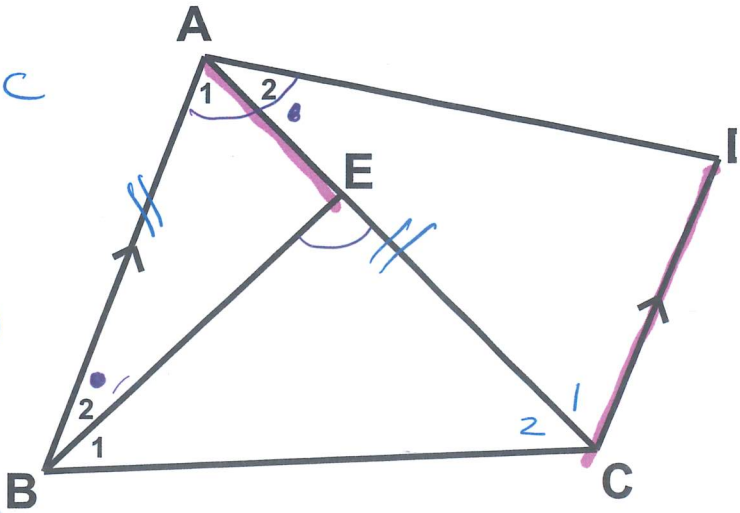
$\Delta ABC \equiv \Delta DCB$ (SAS)

$AC = DB$ (deduced)

QUESTION 9

In the diagram: $AB = AC$ and $DC = AE$
Prove that:

a) $\Delta ACD \equiv \Delta BAE$



1. $\hat{C}_1 = \hat{A}_1$ alt ∠s; AB // DC
2. $AC = AB$ given
3. $DC = AE$ given

$\Delta ACD \equiv \Delta BAE$ (SAS)

b) $B\hat{A}D = B\hat{E}C$

$\hat{B}_2 = \hat{A}_2$ (∠'s of ≡ Δs)

$B\hat{A}D = \hat{A}_1 + \hat{A}_2$

$B\hat{E}C = \hat{A}_1 + \hat{B}_2$ ext ∠ Δ AEB

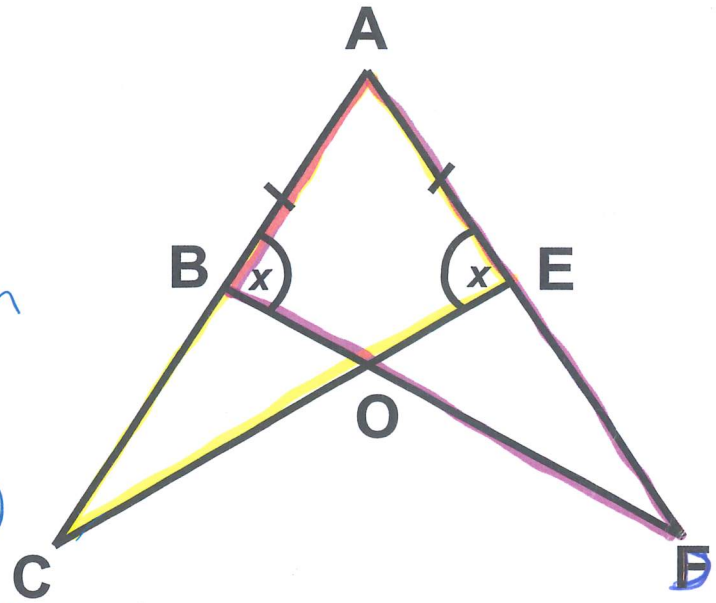
$\hat{A}_1$ is common

$\therefore \Delta BA\hat{D} = B\hat{E}C$

SERVICE BEFORE SELF
S.M.D.S.G.
GOD BEFORE ALL

QUESTION 10

Given $AB = AE$. Prove that $CE = ~~BD~~$. BD



In Δ ABF & Δ AEC

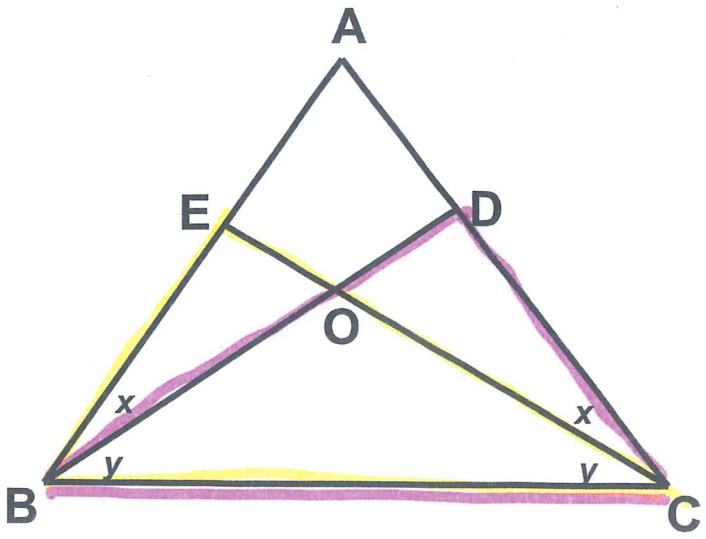
1. $AB = AE$ given
2. $A\hat{B}F = A\hat{E}C = x$ given
3. $\hat{A}$ is common

$\Delta ABF \equiv \Delta AEC$ (SAA)

$CE = BF$ (deduced)

QUESTION 11

Prove that $BD = CE$.



In ΔBDC & ΔCEB

1. $\hat{BCD} = \hat{CBE} = x+y$

2 $\hat{DBC} = \hat{ECB} = y$

3 BC is common

$\Delta BDC \equiv \Delta CEB$ (SAA)

QUESTION 12***

FDCE is a parallelogram. CE is produced to A such that $CE = EA$ and $CD = DB$.
Prove that $\Delta BDF \equiv \Delta FEA$.

In Δ BDF & Δ FEA

1. $\hat{B} = \hat{F}_1$ corresp $\angle$s: FE // BC

2. BD = DC given

DC = FG opp sides parm

BD = FG

3. Let $\hat{E}_2 = \hat{D}_2 = x$ opp $\angle$s of parm

$\therefore \hat{E}_1 = \hat{D}_1 = 180° - x$ adj $\angle$s on str lines

$\Delta BFF \equiv \Delta FEA$ (SAA)

